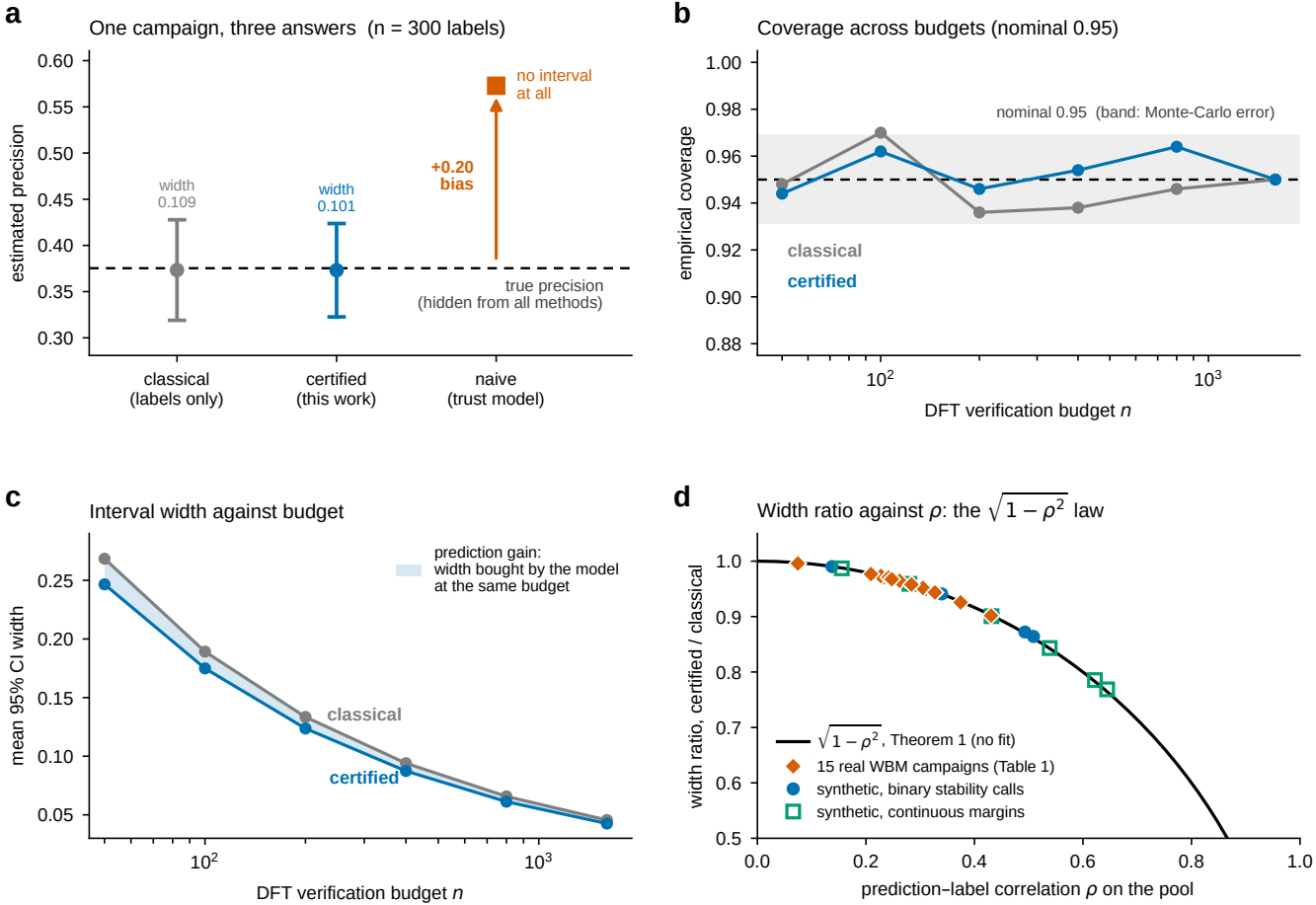
a
One campaign, three answers (n = 300 labels)
estimated precision
0.60
0.55
0.50
0.45
0.40
0.35
0.30
width
0.109
width
0.101
+0.20
bias
no interval
at all
true precision
(hidden from all methods)
classical
(labels only)
certified
(this work)
naive
(trust model)
b
Coverage across budgets (nominal 0.95)
empirical coverage
1.00
0.98
0.96
0.94
0.92
0.90
0.88
nominal 0.95 (band: Monte-Carlo error)
classical
certified
10^2
10^3
DFT verification budget n
c
Interval width against budget
mean 95% CI width
0.25
0.20
0.15
0.10
0.05
prediction gain:
width bought by the model
at the same budget
classical
certified
10^2
10^3
DFT verification budget n
d
Width ratio against ρ: the $\sqrt{1-\rho^2}$ law
width ratio, certified / classical
1.0
0.9
0.8
0.7
0.6
0.5
$\sqrt{1-\rho^2}$, Theorem 1 (no fit)
15 real WBM campaigns (Table 1)
synthetic, binary stability calls
synthetic, continuous margins
0.0
0.2
0.4
0.6
0.8
1.0
prediction–label correlation ρ on the pool